Stanford

Weekly update 2 of 5 · week of Nov 16

Baseline (prompt v1) and error analysis

THIS UPDATE IN THREE NUMBERS

19.5%

of estimates fall within ±20% of the reference LCA

0.75

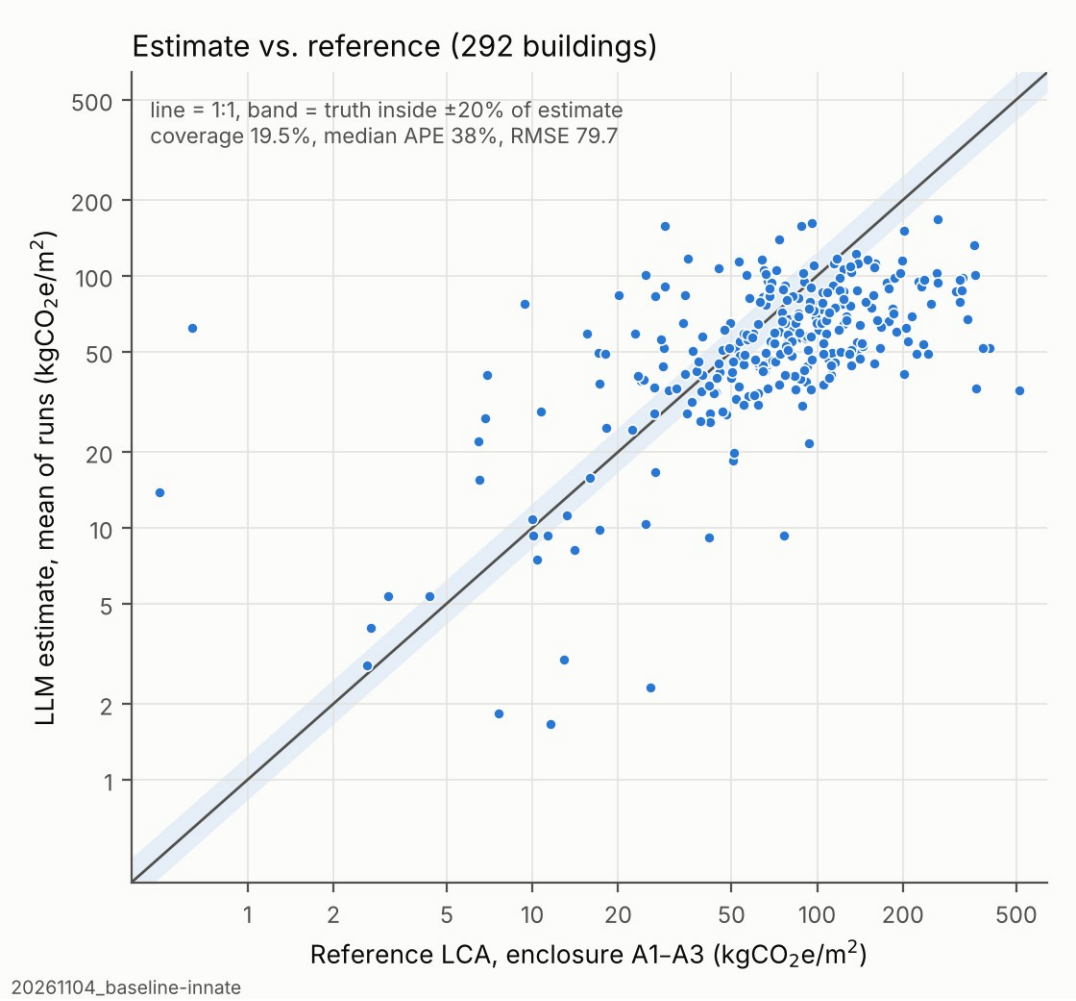
bias: estimates are about 25% low on average

0.46

log–log slope: every building is pulled toward a typical value

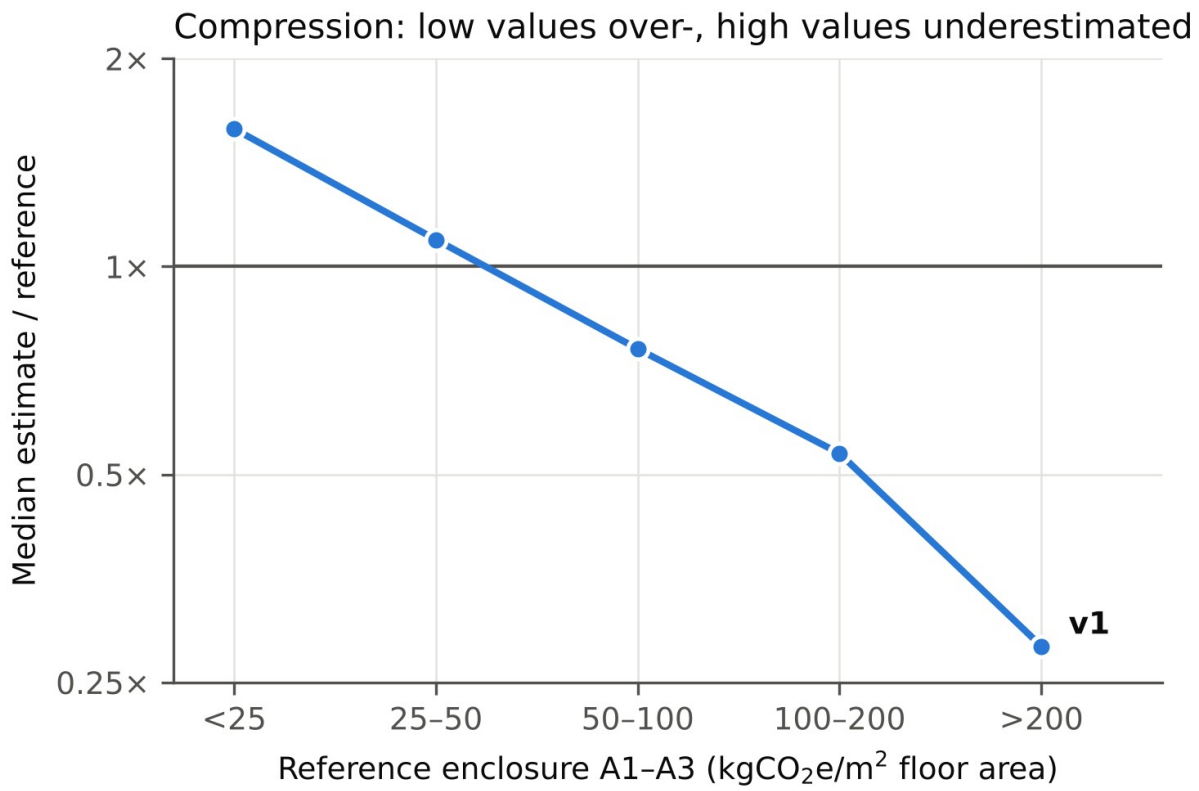
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Baseline: a plain prompt, the model's own knowledge



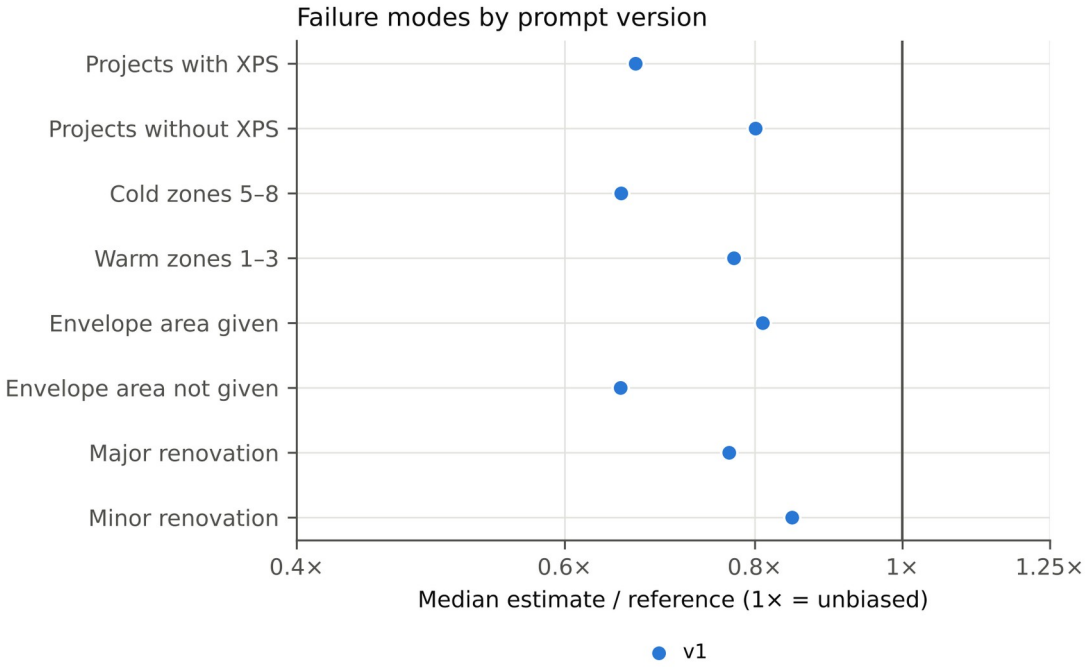
- **Coverage 19.5%** (CI 15.4–24.5%), RMSE 79.7, median APE 37.9%, CV 19.0% across runs.
- **63%** of buildings are underestimated beyond the band; 17.5% are overestimated.
- The **20 largest errors**, all underestimates, carry 68% of the squared error.

Main pattern: compression, not uniform underestimation



- Below 25 $kgCO_2e/m^2$: overestimated **1.58×**. Above 200: estimated at **0.28×**.
- The model leans on **geometry**: the envelope-to-floor ratio correlates 0.62 with the log estimate but only 0.39 with the log reference.
- What it misses is **carbon per m² of envelope** ($r = -0.79$ with the log error). That is set by assembly and material choices, and **no feature describes the wall assembly**.

Why the model underestimates



1. **HFC-blown XPS is never considered:** 0.6% of answers mention XPS. Ratio 0.67 with XPS vs 0.80 without; removing XPS from the reference moves 0.72 → 0.84, about **half the bias**.
2. **The climate gap is an insulation effect:** cold zones 0.65, warm 0.77. XPS is in 65–68% of zone 5A/5B projects but 26–33% of zone 3B/3C.
3. **Missing envelope area:** 0.81 when given, 0.65 when the model must estimate it.
4. **Curtain-wall tail and renovations:** single projects reach 300 from curtain wall alone; 60% of major renovations are underestimated.
5. **Scope wording** was narrower than the dataset's enclosure (11% of enclosure GWP is tagged "Superstructure" in the source tool).

Why the model overestimates

3.09×

median estimate/reference for the 10 incomplete references

10

projects with no fenestration GWP, flagged

Fit-outs

and minor renovations: envelope scope unknown from the features

- The **largest overestimates trace back to incomplete reference LCAs**, not to the model.
- The remaining overestimates are mostly fit-outs and minor renovations, where the extent of envelope work is not described by any feature.
- Flagged projects: 27, 35, 66, 74, 80, 91, 141, 217, 234, 264.

Next update

Coming next

- **Prompt v2:** add the missing knowledge. A scope matching the dataset, a component take-off, named high-carbon materials, and a renovation rule.
- Rerun 3 × 292 and compare **paired** with v1.

Input we'd like

- Feedback on the **error taxonomy** before we revise the prompt.
- Any **domain knowledge** you would add to the prompt?